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MZ 505
Inventory Systems

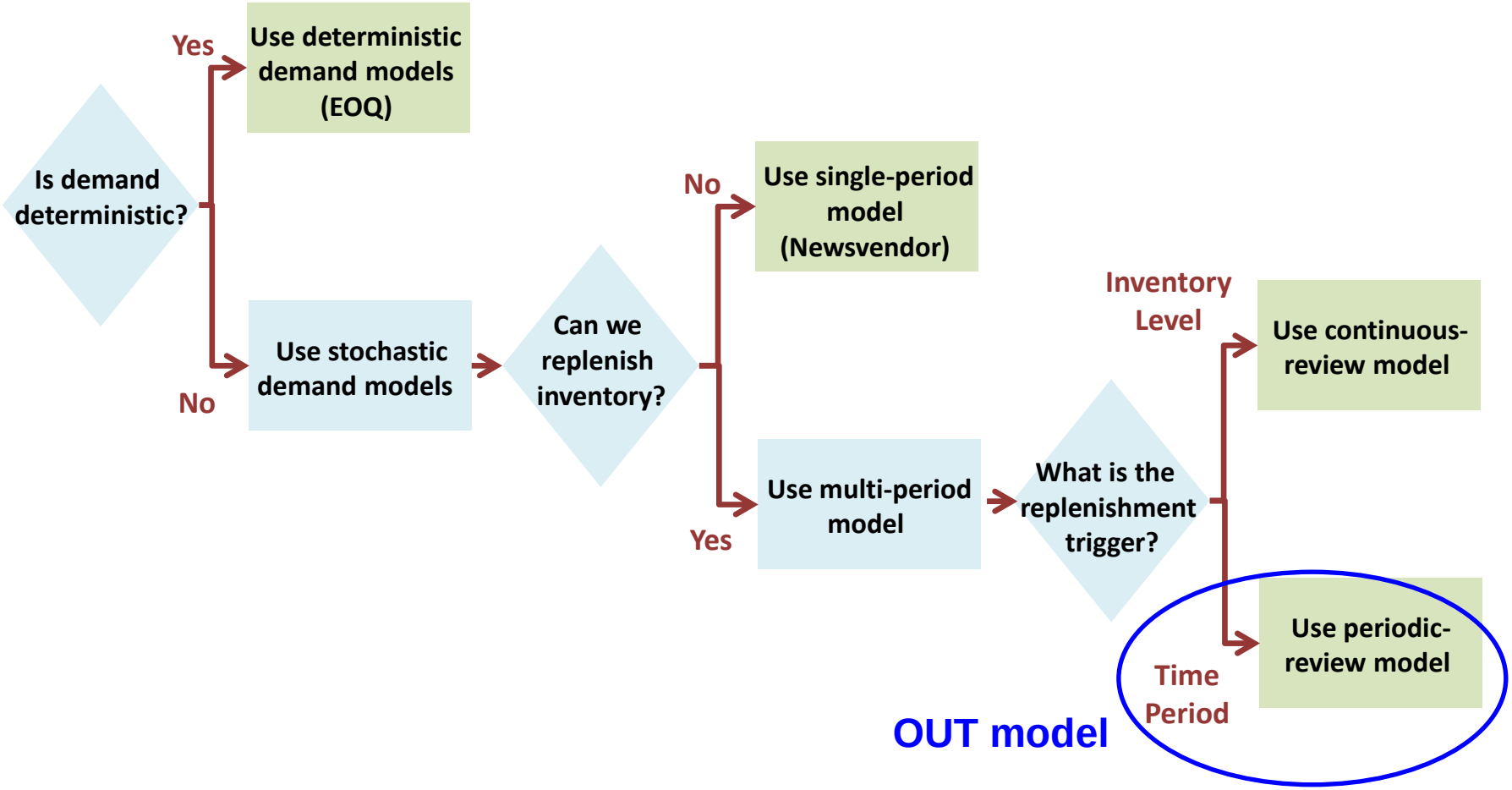
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Session 11: The Order-up-to Inventory Model

Adapted from Cachon and Terwisch slides (2013)

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Decision tree for choosing inventory model



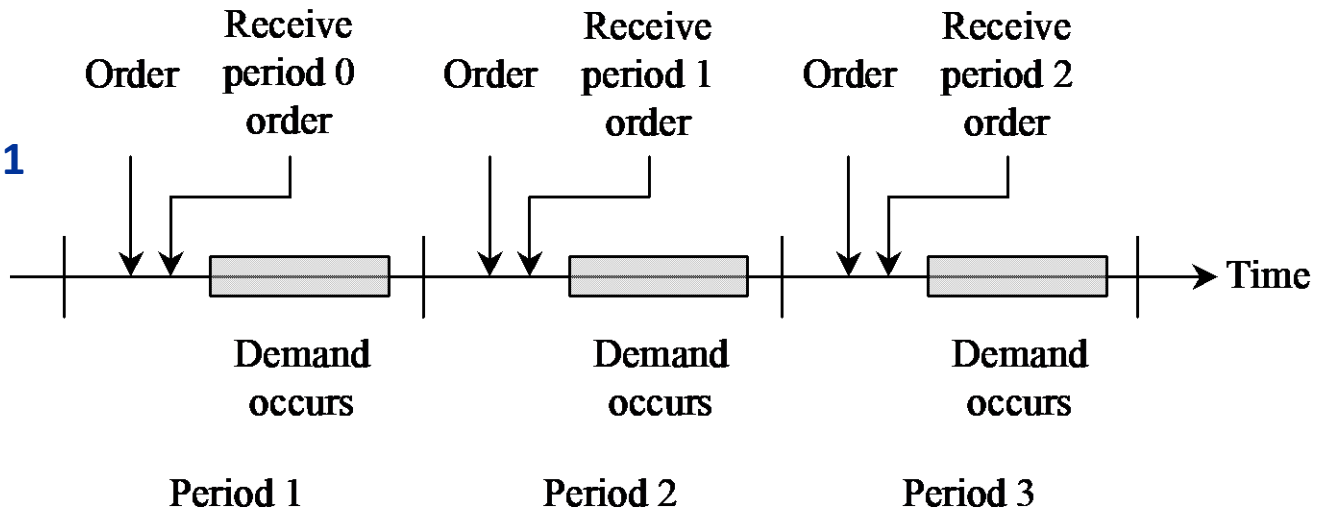
Adapted from Janice H. Hammond, HBP 8016 (2014)

The Order-up-to Model

Timing in the order-up-to model

- Time is divided into periods of equal length, e.g., 1 day, 1 week, etc.
- During a period the following sequence of events occurs:
  - A replenishment order can be submitted
  - Inventory (from some previous order) is received
  - Random demand occurs
- Lead times (L):
  - An order is received after a fixed number of periods (=lead time)

An example with L = 1



Order-up-to model vs. newsvendor model

- Both models have uncertain future demand, but there are differences…

	Newsvendor	Order-up-to
Inventory obsolescence	After one period	Never
Number of replenishments	One (maybe two or three with some reactive capacity)	Unlimited
Demand occurs during replenishment	No	Yes

- Newsvendor applies to short life-cycle products with uncertain demand and the order-up-to applies to long life-cycle products with uncertain, but stable (stationary), demand.

The Order-up-to Model: Assumptions and implementation

Some definitions

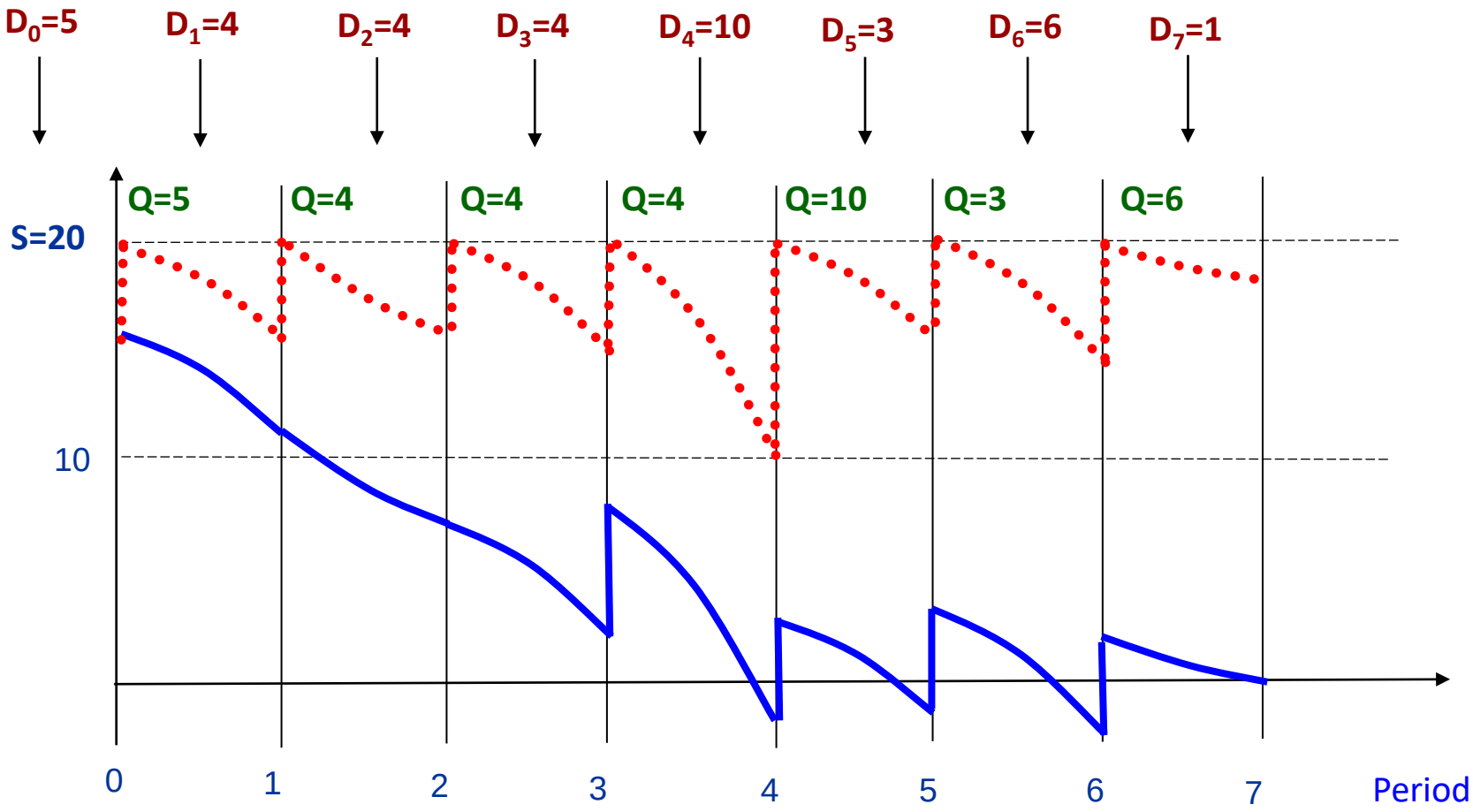
- **On-order (or pipeline) inventory** = the number of units that have been ordered but have not been received.
- **On-hand inventory** = the number of units physically in inventory ready to serve demand.
- **Backorder** = the total amount of demand that has not been satisfied.
  - All backordered demand is eventually filled, i.e., there are no lost sales.
- **Inventory level** = On-hand inventory – Backorder
- **Inventory position** = Inventory level + On-order inventory
- **Order-up-to level, S**
  - The maximum inventory position we allow
  - Sometimes called the **base-stock level**

Order-up-to model implementation

- Each period's **order quantity = S – Inventory position**
  - Suppose S = 4
    - If a period begins with an inventory position = -3, how many units will be ordered? Answer: 7
    - If demand were 10 in period 4, then how many units are ordered at the start of period 5? Answer: 10

- A period's order quantity = the previous period's demand:
  - The order-up-to model is a "pull system" because inventory is ordered in response to demand.
  - The order-up-to model is sometimes referred to as a 1-for-1 ordering policy.

Example: L = 3, S = 20 and Starting Inventory (at t_0) = 15



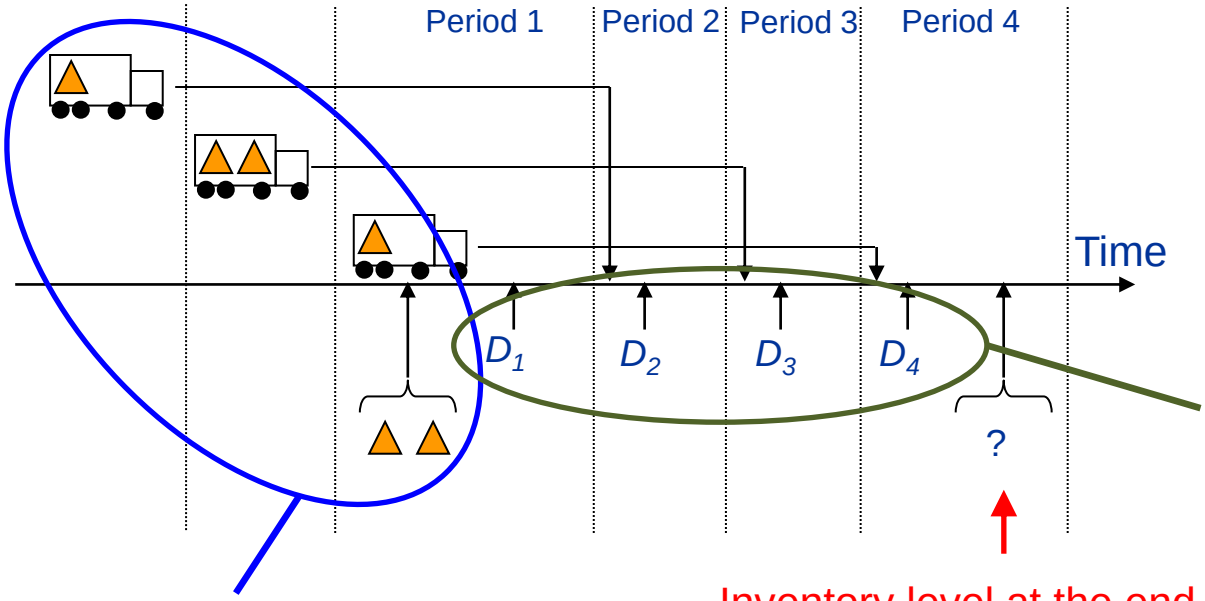
Inventory position

Inventory level

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Inventory level at the end of a period?

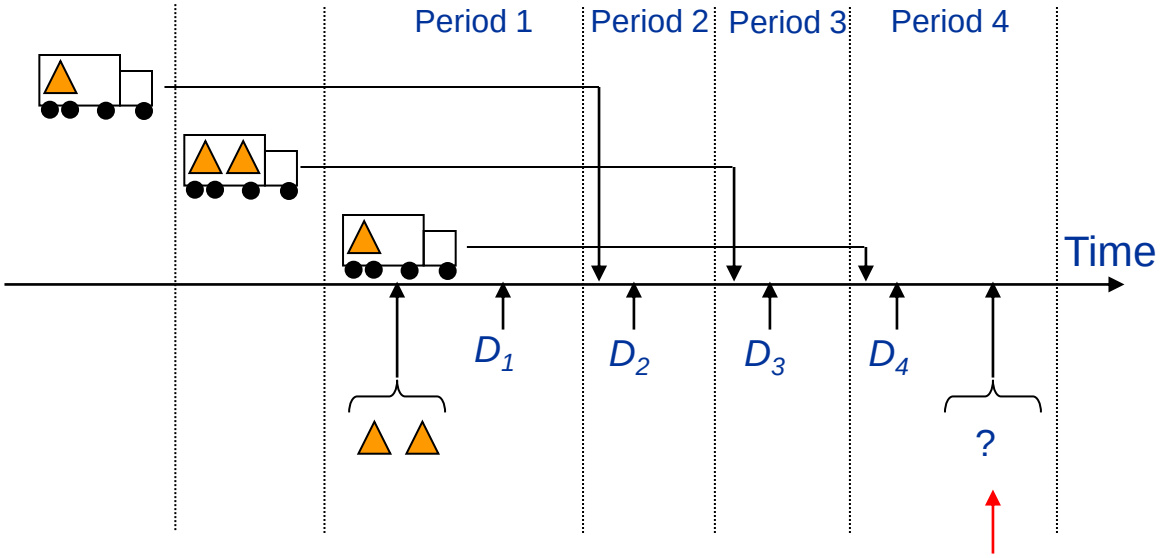
- What determines the inventory level at the end of a period?
  - Inventory level at the end of a period = S – (Demand over L+1 periods)
- Explanation via an example with *S* = 6, L = 3, and 2 units on-hand at the start of period 1



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Inventory level at the end of a period?

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- Explanation via an example with S = 6, L = 3, and 2 units on-hand at the start of period 1



Inventory level at the end of period four
= 6 - D_1 – D_2 – D_3 – D_4

Keep in mind:

At the start of a period (after our order is placed), we have Inventory level + On-order = S

All inventory on-order at the start of period 1 arrives before the end of period 4

Nothing ordered in periods 2-4 arrives by the end of period 4

All demand is satisfied so there are no lost sales

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The Order-up-to Model:

Choosing an order-up-to level, S, to meet an in-stock service target

Calculating demand over L+1 periods: Normally distributed demand

- Assume that the period length is **1 week**, the lead time is **L weeks**, and each month has 4 weeks.
- Assume that **monthly demand** is normally distributed with mean and standard deviation equal to **μ** and **σ**, respectively.
- Then,
  - Weekly demand is normally distributed with…
    - Mean $= \mu/4$
    - Standard deviation $= \frac{\sigma}{\sqrt{4}} = \frac{\sigma}{2}$
  - L+1 demand is normally distributed with…
    - Mean $= \frac{\mu}{4} \times (L + 1)$
    - Standard deviation $= \frac{\sigma}{2} \times \sqrt{L + 1}$

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Calculating demand over L+1 periods: Poisson distributed demand

- Assume that the period length is 1 day and the lead time is L days.
- Assume that daily demand is Poisson with mean equal to **λ**
- Then,
  - L+1 demand is Poisson with…
    - Mean $= \lambda \times (L + 1)$

Choosing S to meet a target in-stock with normally distributed demand

- Suppose the **target in-stock probability** is 99.9%; that is, Prob{Demand over (L+1) periods ≤ S} = 0.999 :
  - From the Standard Normal Distribution Function Table, F(3.08)=0.9990
  - So we choose z = 3.08
  - To convert z into an order up-to level:

$$S = \mu_{L+1} + z \cdot \sigma_{L+1}$$

  - Note that μ and σ are the parameters of the normal distribution that describes demand over L+1 periods.

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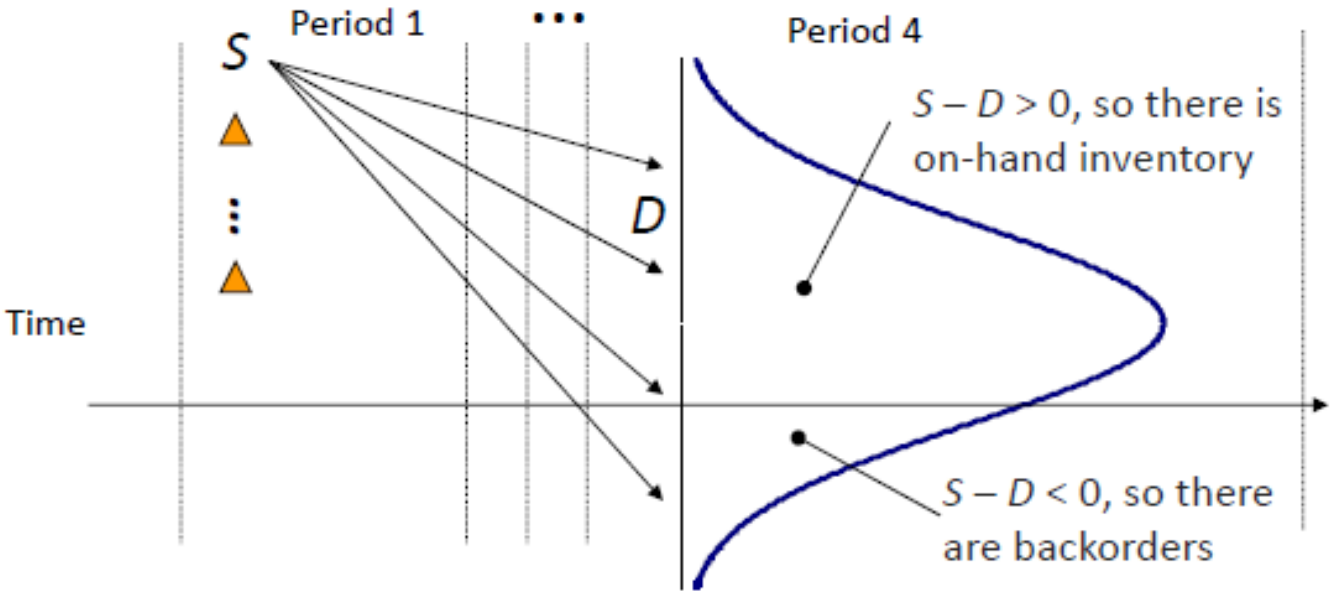
The Order-up-to Model: Performance measures

Stockout and in-stock probabilities

- The **in-stock probability** is the probability all demand is filled in a period:

  In-stock probability = Prob{Demand over (L+1) periods ≤ S}

- The **stockout probability** is the probability at least one unit is backordered in a period.



D = demand over L+1 periods

Expected backorder for a given order-up-to level, S

- **Expected backorder** is analogous to the Expected lost sales in the Newsvendor model. When demand (per period) is Normal:
  - Normalize the order-up-to level S:

$$z = \frac{S - \mu_{L+1}}{\sigma_{L+1}}$$

  - Lookup L(z) in the Standard Normal Loss Function Table
  - Convert expected loss, L(z), for the standard normal into the expected backorder with the actual normal distribution that represents demand over L+1 periods:

$$E[\text{backorder}] = \sigma_{L+1} L(z)$$

  - Therefore, if the order-up-to is S, then on average there are $\sigma_{L+1} L(z)$ backorders at the <u>end of any period.</u>

Other performance measures given S

- **Expected on-hand inventory (or "Safety Stock")** is analogous to Expected left-over inventory in the Newsvendor model:

$$\text{E[on-hand]} = S - \text{E}[D_{L+1}] + \text{E[backorder]}$$

  - This is the expected on-hand inventory at the end of a period.

- **Expected on-order inventory** is the average number of units on order at any given time:

$$\text{E[on-order]} = \text{Lead Time} \times \text{E[Demand in one period]}$$

  - Due to Little's Law, expected on-order inventory does not depend on demand variability (note that it equals the expected demand over L periods, not L+1 periods)

- Question: How much is the **Expected cycle inventory**? Answer: μ/2

Other performance measures given S (cont.)

- **Fill rate (or expected fill rate)** is the fraction of customer demand per period satisfied from product in inventory:

$$\text{Fill Rate} = \frac{\text{E[demand satisfied from inventory]}}{\text{E[demand in one period]}} = 1 - \frac{\text{E[backorder]}}{\mu}$$

  - Stockout occurs when the demand during L+1 periods exceeds the order-up-to level, S.
  - Note that while D_{L+1} is the basis for the calculation of E[backorder], the denominator in the fill rate considers the demand per period.

The Order-up-to Model:

Appropriate service levels

What does a 99.9% in-stock target mean?

- Let "α" be the in-stock probability target for a period
- Let N be the number of periods in an interval of interest (e.g., one month, one year, etc.)
- The expected number of periods in which an out-of-stock occurs = N x (1-α)
- The expected number of periods before the first out-of-stock occurs = 1 / (1-α)

	Expected time to 1st day out of stock*		Expected number of days out of stock per year*		
In-stock	Days	Years	per year	per 3 years	per 5 years
99.00%	100	0.27	3.7	11.0	18.3
99.50%	200	0.55	1.8	5.5	9.1
99.90%	1000	2.74	0.4	1.1	1.8
99.95%	2000	5.48	0.2	0.5	0.9
99.99%	10000	27.40	0.0	0.1	0.2

* one day = one period

Justifying a service level via cost minimization

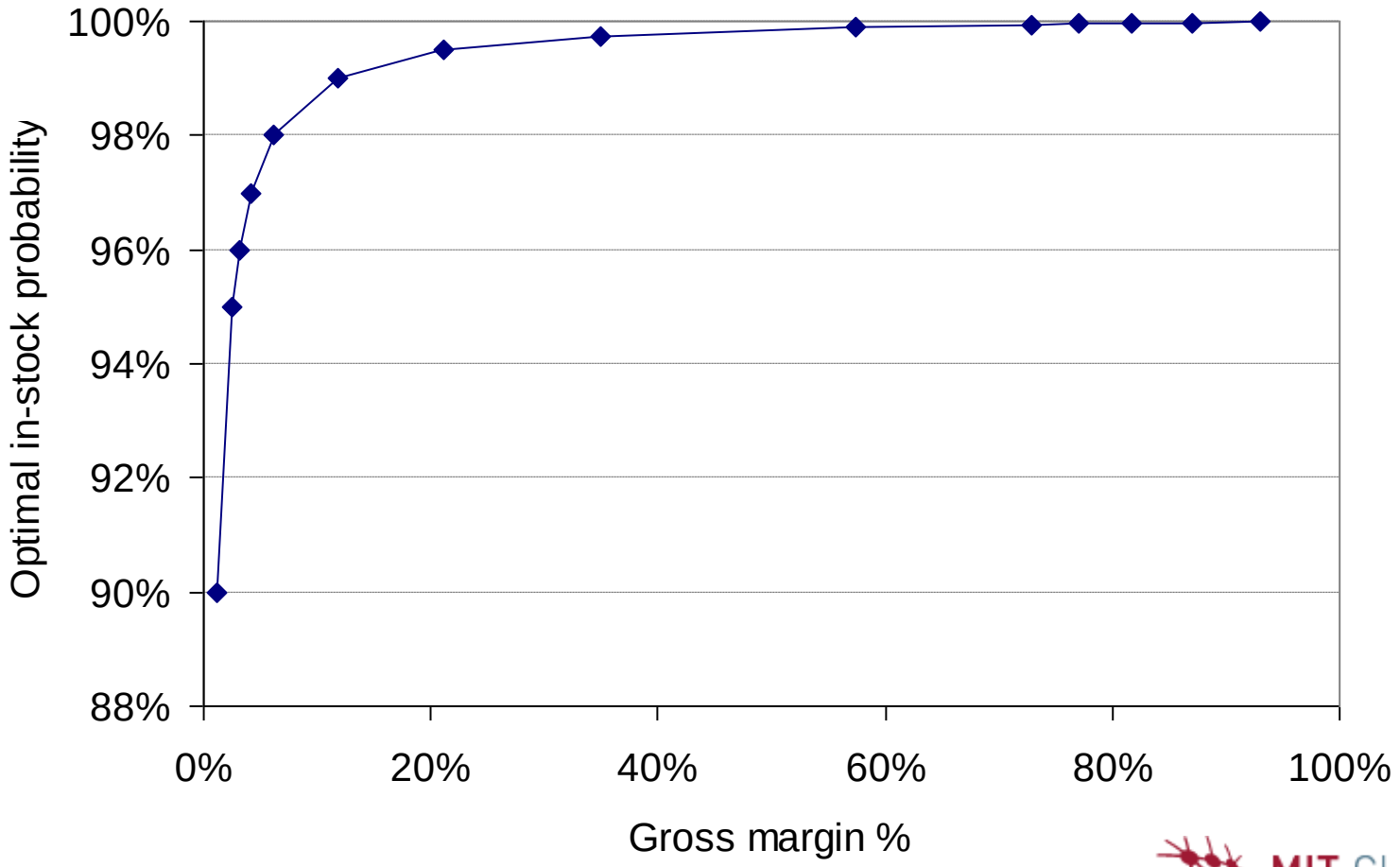
- Let h equal the holding cost per unit per period
- Let b equal the penalty per unit backordered
- "Too much-too little" challenge:
  - If S is too high, then there are holding costs, $C_o = h$
  - If S is too low, then there are backorders, $C_u = b$
- Cost minimizing order up-to level satisfies

$$\text{Prob}\{D_{L+1} \leq S\} = \frac{C_u}{C_u + C_o} = \frac{b}{b+h} \quad \text{(critical ratio)}$$

  - Suppose the review period is one week, the holding cost per unit per week is \$0.5 and the cost (penalty) per unit backordered is \$20.
  - Optimal in-stock probability is 20/(20+0.5)=97.56%

The optimal in-stock probability is usually quite high

- Suppose the annual holding cost is 35% and the backorder penalty cost equals the gross margin



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Readings

- Chapter 14 from Cachon & Terwiesch, 3rd edition

Thank you!